

CS401: Introduction to Neural Networks & Deep Learning

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1. Biological Inspiration & The Artificial Perceptron

Artificial Neural Networks (ANNs) are computational systems inspired by biological brains. The fundamental computational building block is the artificial neuron (perceptron). A perceptron takes multiple weighted inputs, adds a learnable bias term, and passes the result through an activation function to generate an output scalar.

2. Critical Role of Non-Linear Activation Functions

Without non-linear activation functions, a multi-layer neural network collapses into a single linear matrix transformation, regardless of depth. Key activation functions discussed in this lecture include:

- Sigmoid Function: Squashes input values between 0 and 1, traditionally used for binary probability.
- ReLU - Rectified Linear Unit: Computationally efficient, accelerates convergence, and avoids gradient vanishing.
- Softmax: Normalizes a K-dimensional vector of raw logits into a valid probability distribution where probabilities sum to 1.0.

3. Optimization via Backpropagation & Gradient Descent

Neural networks learn optimal weights by minimizing an empirical loss function. Backpropagation leverages the chain rule of differential calculus to efficiently compute partial derivatives of loss with respect to all layer weights. Gradient descent iteratively updates parameters in the opposite direction of the gradient vector: $W_{\text{new}} = W_{\text{old}} - \text{learning_rate} * dL/dW$.

4. Overfitting & Regularization Strategies

High model capacity can lead to overfitting on training data. Common mitigation techniques include Dropout (randomly deactivating neurons during forward pass) and L2 Weight Decay.